TEXAS SALARY PREDICTION

PRCP-1024 — Complete End-to-End Project Workflow

1. Project Objective

Build one complete Jupyter Notebook that analyzes Texas state government salary data, answers the required business questions, develops a predictive model, compares multiple models, identifies the best model for production, and documents challenges and solutions.

2. Required Business Questions

• Who are the outliers in the salaries?

• What departments/roles have the biggest wage disparities between managers and employees?

• Have salaries and total compensations for some roles/departments/head-count changed over time?

3. Complete Workflow

Step	Stage	What to do
01	Project Introduction	Title, objective, problem statement, business questions, dataset source.
02	Import Libraries	pandas, numpy, matplotlib/seaborn, sklearn and other libraries required by the notebook.
03	Load Dataset	Read the salary dataset and create the initial DataFrame.
04	Data Understanding	Inspect shape, columns, data types, sample records, unique values and descriptive statistics.
05	Data Quality Check	Check missing values, duplicates, invalid values and inconsistent formats.
06	Data Cleaning	Handle missing data, duplicates, incorrect types, salary/date formatting and other data-quality issues.
07	Feature Preparation	Prepare numerical and categorical variables and derive useful features such as employment year where appro
08	Exploratory Data Analysis	Study salary distributions, agencies, roles/class titles, gender, ethnicity and employee status.
09	Outlier Analysis	Use suitable statistical/visual methods to identify unusually high or low salaries and investigate the affected re
10	Wage Disparity Analysis	Compare manager and employee salaries by relevant department/role and calculate salary gaps.
11	Time Analysis	Analyze salary, total compensation and head-count changes over time using the employment-date information
12	Machine Learning Dataset	Define Annual as the target variable and prepare suitable input features.
13	Train/Test Split	Split the prepared data into training and testing data.
14	Model Training	Train multiple regression models suitable for annual salary prediction.
15	Model Evaluation	Evaluate models using R^2, MAE and RMSE or other appropriate regression metrics.
16	Model Comparison	Create a comparison table and identify the strongest model based on the evaluation results.
17	Final Prediction	Use the selected model to demonstrate annual salary prediction.
18	Challenges & Solutions	Document data/model challenges, the techniques used and the reasons for choosing them.
19	Final Findings	Summarize important salary, outlier, disparity, time and modeling findings.
20	Conclusion	State the final business conclusion and recommended production model.

4. Recommended Notebook Structure

• 01. Project Title

• 02. Problem Statement

• 03. Business Questions

• 04. Import Libraries

• 05. Load Dataset

• 06. Data Understanding

• 07. Data Cleaning

• 08. Exploratory Data Analysis

• 09. Outlier Analysis

• 10. Manager vs Employee Wage Disparity

• 11. Time-Based Analysis

• 12. Feature Engineering

• 13. Feature Selection

• 14. Train-Test Split

• 15. Model 1

• 16. Model 2

• 17. Model 3

• 18. Model Comparison Report

• 19. Best Model

• 20. Salary Prediction

• 21. Challenges and Solutions

• 22. Final Findings

• 23. Conclusion

5. Data Columns to Work With

Agency, Agency Name, Last Name, First Name, MI (Middle Initial), Class title, Ethnicity, Gender, Status, Employ Date, Hourly rate, Hrs per week, Monthly (Monthly income), Annual (Annual Income), State number.

6. Machine Learning Flow

1. Select Annual as the prediction target.

2. Choose relevant employee/job features as predictors.

3. Clean and encode categorical features.

4. Scale features only when required by the chosen model.

5. Split the data into training and testing sets.

6. Train multiple regression models.

7. Calculate R^2, MAE and RMSE.

8. Compare the results.

9. Select the best-performing model.

10. Use the selected model for final salary prediction.

7. Analysis Outputs to Include

• Dataset shape and sample records

• Missing-value and duplicate summary

• Descriptive statistics

• Salary distribution chart

• Average salary by agency

• Salary by class title/role

• Salary by gender and ethnicity where appropriate

• Outlier visualization and outlier records

• Manager-versus-employee wage-gap analysis

• Salary/compensation/head-count trend over time

• Model performance comparison table

• Best-model result

• Challenges and solutions

• Final conclusion and recommendations

8. Model Comparison Template

Model	R^2	MAE	RMSE	Decision
Linear Regression	—	—	—	Evaluate
Random Forest	—	—	—	Evaluate
Gradient Boosting	—	—	—	Evaluate
Other suitable model	—	—	—	Evaluate
Best Model	—	—	—	Select after testing

9. Challenges & Solutions Template

Challenge	Technique/Solution	Reason
Missing values	To be filled after inspection	Based on the actual missing-data pattern
Categorical variables	Encoding technique	Required for ML models
Outliers	Statistical + business investigation	Avoid removing genuine salary records blindly
Large/high-cardinality categories	Feature handling/selection	Reduce unnecessary model complexity
Model performance	Compare multiple models	Select the strongest production candidate

Important: Do not invent final findings or model scores before running the notebook. Fill the result sections using the actual dataset and actual model outputs.

Source basis: PRCP-1024-TexasSalaryPrediction project document. The source requires the analysis, predictive model, business-question analysis, model comparison, and challenges report to be completed in a single Jupyter Notebook.